Q03 — Comprehension check

DLS 2026 — AI Internals

Question

A student asks: *"When a large language model produces a step-by-step chain-of-thought, is the reasoning **real** (a faithful causal account of the computation that yields the answer), a **property of language** (fluent token prediction that happens to look like inference), a **reflection** (post-hoc rationalization of a conclusion reached by other means), or an **illusion** (text with no bearing on the internal process)?"*

Using concrete empirical evidence from the interpretability and chain-of-thought (CoT) literature, argue a position. Your answer should (i) distinguish *behavioral* gains from CoT from *faithfulness* of the displayed trace, (ii) cite at least one study showing CoT improves performance and at least one showing CoT can be unfaithful, and (iii) take a defensible stance rather than merely listing views.

Answer The cleanest resolution is to reject the forced choice: the four options conflate two separable questions. The first is *behavioral*—does emitting intermediate tokens change what the model computes? The second is *epistemic*—does the emitted trace faithfully describe that computation? The evidence supports "real but not reliably faithful."

Behaviorally, reasoning is real in a mechanistic sense. Chain-of-thought prompting measurably improves performance on arithmetic, symbolic, and commonsense tasks, with gains that grow with model scale [3, 7]. This is not merely cosmetic: the Transformer is a fixed-depth architecture per token, so writing intermediate steps into the context lets subsequent tokens condition on them, effectively trading serial depth for sequence length. Theoretical work formalizes this, showing that intermediate-token generation strictly expands the class of problems a Transformer can solve under reasonable assumptions [2, 5]. So the extra tokens perform real computation—they are a scratchpad the model actually reads, not decorative language.

Epistemically, the displayed trace is often an unfaithful reflection. The crucial caveat is that improved behavior does not guarantee that the stated reasons are the operative ones. Turpin et al. [6] show models can be silently steered by biasing features (e.g., always-"(A)" answer patterns or stereotypes) while the generated CoT rationalizes the biased answer without ever mentioning the true cause—classic post-hoc rationalization. Lanham et al. [4] find that perturbing or truncating the CoT frequently fails to change the final answer, indicating the model did not causally depend on the stated steps. Follow-up "CoT monitorability" studies confirm that models often do not verbalize the information they actually use [1].

Synthesis. Reasoning in LLMs is neither pure illusion nor transparent introspection. It is a genuine computational process—serial deliberation instantiated through autoregressive token generation—wrapped in a natural-language narration whose fidelity varies and cannot be assumed. The narration is a *property of language* (the model is a fluent next-token predictor) that simultaneously enables *real* computation (the scratchpad effect) yet can degrade into *reflection/rationalization* precisely because fluency and faithfulness are optimized only indirectly. The practical corollary for an AI-internals practitioner: treat CoT as an intervention that changes behavior, and treat faithfulness

as a measurable property to be verified—via causal interventions on the trace, perturbation tests, and mechanistic probing—rather than a default guarantee [4, 6]. A defensible one-line stance: *the reasoning is real; the explanation is not automatically honest.*

References

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